

A computational framework for empirical Bayes inference

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Abstract In empirical Bayes inference one is typically interested in sampling from the posterior distribution of a parameter with a hyper-parameter set to its maximum likelihood estimate. This is often problematic particularly when the likelihood function of the hyper-parameter is not available in closed form and the posterior distribution is intractable. Previous works have dealt with this problem using a multi-step approach based on the EM algorithm and Markov Chain Monte Carlo (MCMC). We propose a framework based on recent developments in adaptive MCMC, where this problem is addressed more efficiently using a single Monte Carlo run. We discuss the convergence of the algorithm and its connection with the EM algorithm. We apply our algorithm to the Bayesian Lasso of Park and Casella (J. Am. Stat. Assoc. 103:681–686, 2008) and on the empirical Bayes variable selection of George and Foster (J. Am. Stat. Assoc. 87:731–747, 2000).

Keywords Empirical Bayes · Adaptive MCMC · Variable selection · Bayesian LASSO

1 Introduction

This paper develops an adaptive Monte Carlo strategy for sampling from posterior distributions in empirical Bayes (EB) analysis. We start here with a general description of the problem. Suppose that we observe a data $y \in \mathcal{Y}$ generated from a statistical model $f_{\theta, \lambda}(y)$. We take a Bayesian viewpoint and assume that $f_{\theta, \lambda}(y)$ is the conditional distribution

of y given that the parameter takes value $(\theta, \lambda) \in \Theta \times \Lambda$. We treat λ as a hyper-parameter and assume that the conditional distribution of the parameter θ given $\lambda \in \Lambda$ is $\pi(\theta|\lambda)$. We assume also that the spaces $\mathcal{Y}$ and Θ are equipped with appropriate measure-theoretical structures and that Λ is an open subspace of the n_λ -dimensional Euclidean space $\mathbb{R}^{n_\lambda}$. The joint distribution of (y, θ) given λ is thus

$$\pi(y, \theta|\lambda) = f_{\theta, \lambda}(y)\pi(\theta|\lambda).$$

The posterior distribution of θ given y, λ is then given by

$$\pi(\theta|y, \lambda) = \frac{\pi(y, \theta|\lambda)}{\pi(y|\lambda)}, \quad (1)$$

where $\pi(y|\lambda) \stackrel{\text{def}}{=} \int \pi(y, \theta|\lambda) d\theta$ is the marginal distribution of y given λ .

We can handle the hyper-parameter λ in two ways. In a fully Bayesian framework, a prior distribution $\pi(\lambda)$ is assumed for λ and instead of (1), one is interested in sampling from the posterior distribution of θ given y

$$\pi(\theta|y) = \int \pi(\theta|y, \lambda) \omega(\lambda|y) d\lambda, \quad (2)$$

where $\omega(\lambda|y) \propto \pi(y|\lambda)\pi(\lambda)$ is the posterior distribution of λ given y . The main drawback of (2) is that the posterior distribution $\omega(\lambda|y)$ can be sensitive to the choice of the prior $\pi(\lambda)$. Another approach for dealing with λ that has proven very effective in practice is empirical Bayes. The idea consists of using the data y to propose an estimate for λ . This estimate is typically taken as the maximum likelihood estimate $\hat{\lambda}$ of λ given y ,

$$\hat{\lambda} = \text{Argmax } \pi(y|\lambda). \quad (3)$$

Inference about θ is then drawn by sampling from

$$\pi(\theta|y, \hat{\lambda}). \quad (4)$$

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Empirical Bayes inference is particularly useful in hierarchical models where the method can be seen as a Bayesian implementation of Stein's estimator (see e.g. Morris 1983 and Carlin and Louis 2000 for more discussion).

The computation of empirical Bayes estimates involves the following two steps. First, find the maximum likelihood estimate $\hat{\lambda} = \text{Argmax } \pi(y|\lambda)$ and second, sample from distribution $\pi(\theta|y, \hat{\lambda})$ using typically Markov Chain Monte Carlo algorithms. Often, the marginal distribution $\pi(y|\lambda) = \int \pi(y, \theta|\lambda) d\theta$ is not available in closed form, making the maximum likelihood estimation computationally challenging. In these challenging cases, EB procedures can be implemented using the EM algorithm as proposed by Casella (2001). This leads to a two-stage algorithm where in the first stage, a EM algorithm is used (each step of which typically requiring a fully converged MCMC sampling from $\pi(\theta|y, \lambda)$) to find $\hat{\lambda}$ and in a second stage, a Markov Chain Monte Carlo sampler is run to sample from $\pi(\theta|y, \hat{\lambda})$.

This paper proposes a simpler framework where both issues (finding $\hat{\lambda}$ and drawing samples from $\pi(\theta|y, \hat{\lambda})$) are addressed simultaneously, in a single simulation run. It results in a sampler that is computationally more efficient than the EM-followed-by-MCMC. The proposed algorithm is based on stochastic approximation algorithms and builds on recent developments in adaptive Markov Chain Monte Carlo methodology (see e.g. Andrieu and Thoms 2008; Atchade et al. 2009 and the references therein).

The rest of the paper is organized as follows. In Sect. 2, we describe the proposed algorithm and discuss its connection with the EM algorithm and other similar algorithms. How to correct the empirical Bayes inference to account for the fact that the hyper-parameter is estimated is discussed in Sect. 2.3. We analyze the convergence of the algorithm in Sect. 4 and discuss its application to the Bayesian Lasso of Park and Casella (2008) and to the empirical Bayes variable selection of George and Foster (2000).

2 Sampling from empirical Bayes posterior distribution

Continuing with the notations above, we shall use ∇_x to denote the partial derivative with respect to x . Let $\ell(\lambda|y) \stackrel{\text{def}}{=} \log \pi(y|\lambda)$ be the marginal log-likelihood of λ given y and note $h(\lambda|y) \stackrel{\text{def}}{=} \nabla_\lambda \ell(\lambda|y)$ its gradient. Assuming that the interchange of integration and derivation is permissible, we have:

$$\begin{aligned} h(\lambda|y) &= \nabla_\lambda \ell(\lambda|y) \\ &= \int \frac{\partial}{\partial \lambda} \log [f_{\theta, \lambda}(y) \pi(\theta|\lambda)] \pi(\theta|y, \lambda) d\theta \\ &= \int H(\lambda, \theta) \pi(\theta|y, \lambda) d\theta, \end{aligned}$$

where

$$H(\lambda, \theta) \stackrel{\text{def}}{=} \nabla_\lambda \log (f_{\theta, \lambda}(y) \pi(\theta|\lambda)).$$

Notice that in many cases the likelihood does not depend on the hyper-parameter so that the function H simplifies further to

$$H(\lambda, \theta) = \nabla_\lambda \log \pi(\theta|\lambda).$$

We search for $\hat{\lambda} = \text{Argmax } \pi(y|\lambda)$ by solving the equation $h(\lambda|y) = 0$. If h is tractable, this equation can be easily solved analytically or using classical iterative methods. For example the gradient method would yield an iterative algorithm of the form

$$\lambda' = \lambda + ah(\lambda|y),$$

for a step-size $a > 0$. Since, h is typically intractable, we naturally turn to stochastic approximation (SA) algorithms which in essence are stochastic algorithms that mimic the gradient algorithm. Suppose that we have at our disposal for each $\lambda \in \Lambda$, a transition kernel P_λ on Θ with invariant distribution $\pi(\theta|y, \lambda)$. We let $\{a_n, n \geq 0\}$ be a non-increasing sequence of positive numbers such that

$$\lim_{n \rightarrow \infty} a_n = 0 \quad \sum a_n = \infty, \quad \sum a_n^2 < \infty. \quad (5)$$

The stochastic approximation algorithm proposed to maximize the function $\ell(\lambda|y)$ is the following.

Algorithm 2.1 *Initially, we start with $\theta_0 \in \Theta$, $\lambda_0 \in \Lambda$. At time n , Given (θ_n, λ_n) :*

- generate $\theta_{n+1} \sim P_{\lambda_n}(\theta_n, \cdot)$ and*
- calculate $\lambda_{n+1} = \lambda_n + a_n H(\lambda_n, \theta_{n+1})$.*

Stochastic approximation algorithms are well-known algorithms to solve intractable optimization problems. We refer the reader to the monographs of Benveniste et al. (1990) and Kushner and Yin (2003) for a detailed discussion. In Algorithm 2.1, under appropriate conditions, λ_n can be shown to converge to $\hat{\lambda}$ and the marginal distribution of θ_n can be shown to converge in total variation to $\pi(\theta|y, \hat{\lambda})$ (see Sect. 4). The key condition involved in these convergence results is the condition that the sequence $\{\lambda_n, n \geq 0\}$ remains almost surely in a compact set, a property often referred as *stability*. This stability condition is difficult to check in general and often do not hold unless the step-size $\{a_n, n \geq 0\}$ is very carefully chosen. There is an elegant stabilization technique due to Chen and Zhu (1986) and further studied by Andrieu et al. (2005) which, when the Markov kernels have adequate convergence properties, can turn Algorithm 2.1 into a stable algorithm by using a re-projection technique on randomly varying compact sets. We follow this approach

here. Let $\{K_n, n \geq 1\}$ be an increasing sequence of compact subsets of Λ such that $\bigcup K_n = \Lambda$. Let $\Theta_0 \times \Lambda_0 \subset \Theta \times K_0$ and $\Pi : \Theta \times \Lambda \rightarrow \Theta_0 \times \Lambda_0$ an arbitrary (re-projection) function. For example one can set $\Theta_0 \times \Lambda_0 = (\theta_0, \lambda_0)$ for some arbitrary point $(\theta_0, \lambda_0) \in \Theta \times K_0$ and take $\Pi(\theta, \lambda) = (\theta_0, \lambda_0)$. This is the choice made in the examples below.

Algorithm 2.2 Initially, we start with $\theta_0 \in \Theta$, $\lambda_0 \in \Lambda$, $\zeta_0 = 1$ and $\kappa_0 = 0$. At time n , Given $(\theta_n, \lambda_n, \zeta_n, \kappa_n)$:

- Generate $\bar{\theta} \sim P_{\lambda_n}(\theta_n, \cdot)$ and calculate $\bar{\lambda} = \lambda_n + a_{\zeta_n + \kappa_n} H(\lambda_n, \bar{\theta})$.
- If $\bar{\lambda} \in K_{\zeta_n}$ then set $\theta_{n+1} = \bar{\theta}$, $\lambda_{n+1} = \bar{\lambda}$, $\zeta_{n+1} = \zeta_n$, $\kappa_{n+1} = \kappa_n + 1$.
- If $\bar{\lambda} \notin K_{\zeta_n}$ then set $(\theta_{n+1}, \lambda_{n+1}) = \Pi(\theta_n, \lambda_n)$, $\zeta_{n+1} = \zeta_n + 1$, $\kappa_{n+1} = 0$.

To understand the algorithm, note that ζ_n indexes the compact set in use at time n . As long as λ_n remains in K_{ζ_n} , Algorithm 2.2 is similar to Algorithm 2.1. If $\lambda_n \notin K_{\zeta_n}$, we re-initialize the algorithm starting from $\Pi(\theta_n, \lambda_n)$, set the new compact to K_{ζ_n+1} and set the new sequence of step-size to $\{a_{\zeta_n+1+k}, k \geq 0\}$. The index κ_n plays the same role as played by n in Algorithm 2.1. We reset κ_n to zero at each re-initialization.

In the examples below, we use Algorithm 2.2 instead of Algorithm 2.1. Despite its much improved behavior, Algorithm 2.2 remains sensitive to the choice of the step-size $\{a_n, n \geq 0\}$. If the kernels P_λ have good mixing properties (rapid decrease of autocorrelations), the simple choice $a_n = a/n$ for some constant $a > 0$ (usually $a = 1$) works reasonably well. But if the kernels P_λ do not mix well enough, a much slower sequence a_n is preferable; for example $a_n = a/n^\alpha$, $\alpha \in (0.5, 1)$. Typically we will use $\alpha = 0.8$.

2.1 Connection with the EM algorithm

An EM algorithm is developed in Casella (2001) to deal with empirical Bayes inference, particularly to solve the maximization problem in (3). Since $\pi(y|\lambda) = \int \pi(y, \theta|\lambda) d\theta$, the idea is to treat θ as a missing variable. This naturally leads to a EM algorithm to maximize $\ell(\theta|y)$ with a Q function defined as

$$Q(\lambda'|\lambda) = \int \log \pi(y, \theta|\lambda') \pi(\theta|y, \lambda) d\theta.$$

The EM algorithm (Dempster et al. 1977) is an iterative algorithm each iteration of which involves the following two steps. In the first step and given λ_n , we solve the E step by calculating $Q(\lambda|\lambda_n)$. In the M step the function $\lambda \rightarrow Q(\lambda|\lambda_n)$ is maximized to give λ_{n+1} . These two steps are iterated until convergence. In the present context, the main challenge in applying the EM algorithm is the intractability of the Q function. This is typically addressed by using

Monte Carlo methods as in the Monte Carlo EM (MC-EM) of Wei and Tanner (1990). This consists of replacing the exact calculation of $Q(\cdot|\lambda)$ in the E step by a Monte Carlo approximation $\tilde{Q}(\cdot|\lambda)$. This is precisely the approach taken in Casella (2001). But since the distribution $\pi(\theta|y, \lambda)$ is typically intractable, MCMC is used. Thus, each iteration of the EM algorithm in Casella (2001) takes a full Gibbs sampler from $\pi(\theta|y, \lambda)$. An alternative to Wei-Tanner's Monte Carlo EM is the stochastic approximation EM (SA-EM) of Delyon et al. (1999).

We now elaborate on the link between the EM algorithm described above and Algorithm 2.1. Taking the derivative of $Q(\lambda|\lambda_n)$ with respect to λ , we can easily show that

$$\nabla_\lambda Q(\lambda|\lambda_n) = \int H(\lambda, \theta) \pi(\theta|y, \lambda_n) d\theta.$$

If we replace the full maximization of the Q function by one step of the gradient algorithm, the EM would become

$$\begin{aligned} \lambda_{n+1} &= \lambda_n + a \nabla_\lambda Q(\lambda_n|\lambda_n) \\ &= \lambda_n + a \int H(\lambda_n, \theta) \pi(\theta|y, \lambda_n) d\theta. \end{aligned}$$

This recursion is essentially the EM gradient algorithm of Lange (1995). Comparing this with Algorithm 2.1, we see that in Algorithm 2.1, the integral $\int H(\lambda_n, \theta) \pi(\theta|y, \lambda_n) d\theta$ is approximated by $H(\lambda_n, \theta_{n+1})$ where $\theta_{n+1} \sim P_{\lambda_n}(\theta_n, \cdot)$. Thus Algorithm 2.1 is a sort of "approximate" EM algorithm ran on a faster time scale where both the E and the M steps are only approximately implemented.

2.2 Connection with other SA algorithms in the statistical literature

Many algorithms similar to Algorithm 2.1 have been proposed to solve intractable optimization problems in various contexts. For example Younes (1988) has developed a stochastic approximation method for finding maximum likelihood estimates for Gibbs random fields models which is similar to Algorithm 2.1. The same type of algorithm has also appeared in the maximum likelihood estimation of exponential random graphs in social networks (Snijders 2002). Let us briefly describe these related algorithms.

Keeping the notation above, suppose that we are interested in the maximum likelihood estimate of θ given that we observe $y \sim f_\theta(y)$. We assume that $f_\theta(y) = e^{E(y, \theta)} / Z(\theta)$, for some function $E(x, \theta)$, where $Z(\theta) \stackrel{\text{def}}{=} \int e^{E(x, \theta)} dx$ is an intractable normalizing constant. The log-likelihood is given by

$$\ell(\theta|y) = E(y, \theta) - \log \left(\int e^{E(x, \theta)} dx \right).$$

And

$$\begin{aligned} h(\theta|y) &= \nabla_{\theta} \ell(\theta|y) \\ &= \int (\nabla_{\theta} E(y, \theta) - \nabla_{\theta} E(x, \theta)) \frac{e^{E(x, \theta)}}{Z(\theta)} dx \\ &= \mathbb{E}_{\theta} (\nabla_{\theta} E(y, \theta) - \nabla_{\theta} E(X, \theta)), \end{aligned}$$

where $X \sim e^{E(\theta, \cdot)}/Z(\theta)$. Solving $h(\theta|y) = 0$ is clearly intractable in general. But this equation can be solved easily using an algorithm of the sort of Algorithm 2.1. To be more specific, for $\theta \in \Theta$, let $T_{\theta}(x, A)$ be a Markov kernel on $\mathcal{Y}$ with invariant distribution $f_{\theta}(\cdot)$. A SA algorithm to solve $h(\theta|y) = 0$ is the following.

Algorithm 2.3 Initially, we start with $y_0 \in \mathcal{Y}$ and $\theta_0 \in \Theta$. At time n , given (y_n, θ_n) :

- a Generate $y_{n+1} \sim T_{\theta_n}(y_n, \cdot)$ and
- b calculate $\theta_{n+1} = \theta_n + a_n (\nabla_{\theta} E(y, \theta_n) - \nabla_{\theta} E(y_{n+1}, \theta_n))$.

A distinction with Algorithm 2.1 that is worth pointing out is the fact that the key process of interest in Algorithm 2.3 is $\{\theta_n, n \geq 0\}$ which performs the optimization, whereas in Algorithm 2.1, we are mostly interested in $\{\theta_n, n \geq 0\}$ whose marginal distribution is expected to converge to the posterior distribution (1) with λ replace by $\hat{\lambda}$.

Another related work is Gu and Kong (1998) which proposes a stochastic approximation algorithm similar to Algorithm 2.1 to deal with maximum likelihood estimation in incomplete data models. That is, statistical models where the log-likelihood takes the form

$$\ell(\theta|y) = \log \int f_{\theta}(x, y) dx.$$

This type of algorithms are thoroughly discussed in Cappe and Ryden (2005).

2.3 Empirical Bayes confidence intervals

The empirical Bayes inference described in Sect. 2 has an important limitation in the fact that it does not account for the variability in estimating λ . As the result, confidence intervals and other Bayesian credible sets from $\pi(\theta|y, \hat{\lambda})$ will typically be too short with inappropriate coverage probability. This issue has been investigated by many authors (Laird and Louis 1987; Carlin and Gelfand 1990). The computational gain brought by the our methodology makes it possible to implement fairly easily the bootstrap-corrected empirical Bayes confidence interval proposed by Laird and Louis (1987).

To explain how one can improve on the naive EB inference described above, we note that the empirical Bayes posterior distribution $\pi(\theta|y, \hat{\lambda})$ of Sect. 2 was obtained by replacing the posterior distribution $\omega(\lambda|y)$ of λ by the Dirac

measure $\delta_{\hat{\lambda}}$. A more satisfactory solution would be to replace $\omega(\lambda|y)$ by the marginal distribution of the estimator $\hat{\lambda}$. This would account precisely for the variability in estimating λ . Let $\mu_{\hat{\lambda}}$ be the marginal distribution of the estimator $\hat{\lambda}$. We are thus interested in sampling from

$$\pi(\theta|y) = \int \pi(\theta|y, \lambda) \mu_{\hat{\lambda}}(d\lambda).$$

In most applications, $\mu_{\hat{\lambda}}$ is not known and has to be estimated. One solution, originally developed by Laird and Louis (1987) is to approximate $\mu_{\hat{\lambda}}$ by bootstrap. We follow their parametric bootstrap approach which works as follows. Let $\hat{\lambda}$ the maximum likelihood estimate of λ and K the number of bootstrap replicates. Given $\hat{\lambda}$, we generate independently $\theta^{(1)}, \dots, \theta^{(K)}$ from the prior $\pi(\theta|\hat{\lambda})$ and generate independently $y^{(i)} \sim f_{\theta^{(i)}}(\cdot)$. Now let $\hat{\lambda}^{(i)}$ be the maximum likelihood estimate of λ using data $y^{(i)}$ and let $\hat{\mu}_K$ be the discrete measure with probability mass $1/K$ on $\hat{\lambda}^{(i)}$. The bootstrap-corrected empirical Bayes posterior distribution of θ is thus

$$\begin{aligned} \pi_{\text{EB}}(\theta|y) &= \int \pi(\theta|y, \lambda) \hat{\mu}_K(d\lambda) \\ &= \frac{1}{K} \sum_{i=1}^K \pi(\theta|y, \hat{\lambda}^{(i)}). \end{aligned} \quad (6)$$

This leads to the following algorithm.

Algorithm 2.4

Step 1 Run Algorithm 2.2 to estimate $\hat{\lambda}$.

Step 2 Given $\hat{\lambda}$, and for $i = 1, \dots, K$, generate independently $\theta^{(i)} \sim \pi(\cdot|\hat{\lambda})$ and then $y^{(i)}|\theta^{(i)} \sim f_{\theta^{(i)}}(\cdot)$. For $i = 1, \dots, K$, run Algorithm 2.2 to estimate $\hat{\lambda}^{(i)}$ the maximum likelihood estimate of λ using data $y^{(i)}$.

Step 3 Let $\hat{\mu}_K$ be the discrete measure with probability mass $1/K$ on $\hat{\lambda}^{(i)}$. Sample θ from

$$\pi_{\text{EB}}(\theta|y) = \int \pi(\theta|y, \lambda) \hat{\mu}_K(d\lambda) = \frac{1}{K} \sum_{i=1}^K \pi(\theta|y, \hat{\lambda}^{(i)}).$$

3 Examples

3.1 Example: Bayesian LASSO

We illustrate the algorithms above with the Bayesian Lasso of Park and Casella (2008). Consider the linear model

$$y|\beta, \sigma^2, \mu \sim \mathcal{N}(\mu \mathbf{1}_n + X\beta, \sigma^2 I_n), \quad (7)$$

where $y \in \mathbb{R}^n$ is the vector of response, $\mu \in \mathbb{R}$ the overall mean, X a $n \times p$ matrix of regressors, $\sigma^2 \in (0, \infty)$, $\mathbf{1}_n =$

$(1, \dots, 1) \in \mathbb{R}^n$, I_n the n -dimensional identity matrix and $\mathcal{N}(v, \Sigma)$ denotes the Gaussian distribution with mean v and covariance matrix Σ .

The LASSO method as proposed by Tibshirani (1996) is a shrinkage and variable selection method for linear models. The method works by minimizing the usual sum of squared residuals with a bound on the L^1 norm of the solution. More specifically, LASSO estimate β is the model (7) by solving

$$\min_{\beta} \|y - \mu \mathbf{1}_n - X\beta\|^2 + e^\lambda \sum_{j=1}^p |\beta_j|,$$

for some shrinkage parameter $\lambda \in \mathbb{R}$, where $\|\cdot\|$ denotes the Euclidean norm.

The LASSO estimator is now widely used as it typically outperforms the usual least squares methods. It was noted by Tibshirani (1996) that the LASSO estimate can also be obtained as the posterior mode in a Bayesian analysis of (7) with a double-exponential prior distribution on β . This idea has been exploited among others by Park and Casella (2008) which proposes Bayesian LASSO. The LASSO literature relies on cross-validation methods to select the shrinkage parameter λ . In Bayesian LASSO, Park and Casella (2008) proposes an empirical Bayes approach. In this example, we show that Algorithm 2.1 provides a nice computational framework for the empirical Bayes inference of Bayesian LASSO.

The double-exponential distribution admits a representation as a mixture of Gaussian distributions with exponential mixing density (West 1987)

$$\frac{a}{2} e^{-a|z|} = \int_0^\infty \frac{1}{\sqrt{2\pi}s} e^{-\frac{1}{2s}z^2} \frac{a^2}{2} e^{-as/2} ds, \quad z \in \mathbb{R}, a > 0.$$

Using this representation, and assuming an inverse-Gamma $IG(g_1, g_2)$ prior distribution for σ^2 , and following Park and Casella (2008), we propose a hierarchical prior distribution for (μ, β, σ^2) .

$$\beta | \sigma^2, \tau_1^2, \dots, \tau_p^2 \sim \mathcal{N}(0, \sigma^2 D(\tau^2)),$$

$$D(\tau^2) := \text{diag}(\tau_1^2, \dots, \tau_p^2);$$

$$\pi(\sigma^2, \tau_1^2, \dots, \tau_p^2 | \lambda)$$

$$\propto \left(\frac{1}{\sigma^2} \right)^{g_1+1} e^{-g_2/\sigma^2} \mathbf{1}_{(0,\infty)}(\sigma^2) \\ \times \prod_{j=1}^p \frac{e^{2\lambda}}{2} e^{-e^{2\lambda}\tau_j^2/2} \mathbf{1}_{(0,\infty)}(\tau_j^2),$$

for some hyper-parameter λ which corresponds to the shrinkage parameter. We set $g_1 = 2.01$ and $g_2 = 1$. The parameter μ is given a flat prior. Integrating out μ from the

posterior distribution, one gets

$$\pi(\beta, \sigma^2, \tau_1^2, \dots, \tau_p^2 | y, \lambda) \\ \propto e^{2p\lambda} \left(\frac{1}{\sigma^2} \right)^{g_1 + \frac{n-1}{2} + \frac{p}{2} + 1} \\ \times \exp\left(-\frac{1}{2\sigma^2} \|\bar{y} - X\beta\|^2 - \frac{g_2}{\sigma^2} \right) \\ \times \prod_{j=1}^p \frac{1}{\tau_j} \exp\left[-\frac{1}{2} \left(\frac{\beta_j^2}{\sigma^2 \tau_j^2} + e^{2\lambda} \tau_j^2 \right) \right]. \quad (8)$$

In expression (8) $\bar{y} = y - n^{-1} \sum_{i=1}^n y_i$.

Let $\theta = (\beta, \sigma^2, \tau_1, \dots, \tau_p)$. Given, λ , a Gibbs sampler can be easily implemented to sample from the posterior distribution $\pi(\theta | y, \lambda)$, we refer the reader to Park and Casella (2008) for details. Let us denote K_λ the Gibbs sampler Markov kernel with invariant distribution $\pi(\theta | y, \lambda)$. We adopt an empirical Bayes framework and would like to sample from $\pi(\theta | y, \lambda_\star)$, where $\lambda_\star$ is found by maximum likelihood. In the present case the function $H(\theta, \lambda) = \nabla_\lambda \log \pi(\theta | \lambda)$ takes the form

$$H(\lambda, \theta) = 2p - e^{2\lambda} \sum_{j=1}^p \tau_j^2.$$

Algorithm 2.1 then becomes

Algorithm 3.1

- (i) Initialize $\lambda_0 = 0$, $\theta_0 = (\beta_0, \sigma_0^2, \tau_0^2)$ where $\tau_0^2 = (\tau_{1,0}^2, \dots, \tau_{p,0}^2)$.
- (ii) After the n -th iteration and given λ_n and $\theta_n = (\beta_n, \sigma_n^2, \tau_n^2)$:
 1. generate $\theta_{n+1} \sim K_{\lambda_n}(\theta_n, \cdot)$, where K_λ the Markov kernel of the Gibbs sampler with invariant distribution $\pi(\cdot | y, \lambda)$.
 2. set

$$\lambda_{n+1} = \lambda_n + a_n \left(2p - e^{2\lambda_n} \sum_{j=1}^p \tau_{j,n+1}^2 \right).$$

Alternatively and as explained above an EM algorithm can be used. This is the approach taken by Park and Casella (2008). The Q function takes the form

$$Q(\lambda | \lambda_n) = 2p\lambda - \frac{1}{2} e^{2\lambda} \sum_{j=1}^p \int \tau_j^2 \pi(\theta | y, \lambda_n) d\theta,$$

which yields

$$\lambda_{n+1} = \frac{1}{2} \log \left\{ \frac{2p}{\sum_{j=1}^p \int \tau_j^2 \pi(\theta | y, \lambda_n) d\theta} \right\}.$$

Fig. 1 Diabetes data set. (a) e^{λ_n} from the EM algorithm, (b) e^{λ_n} from Algorithm 3.1, (c) marginal density of $e^{\hat{\lambda}}$ estimated by bootstrap

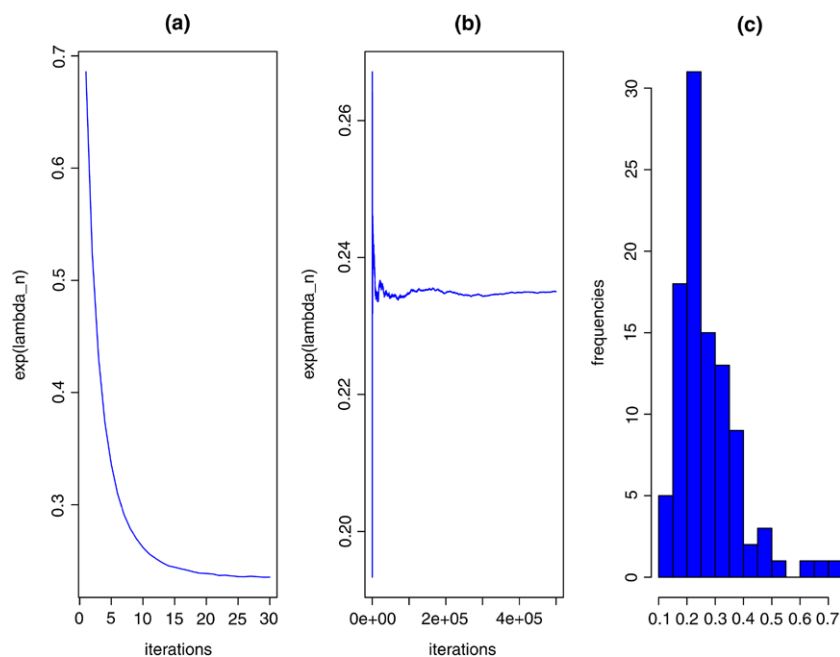
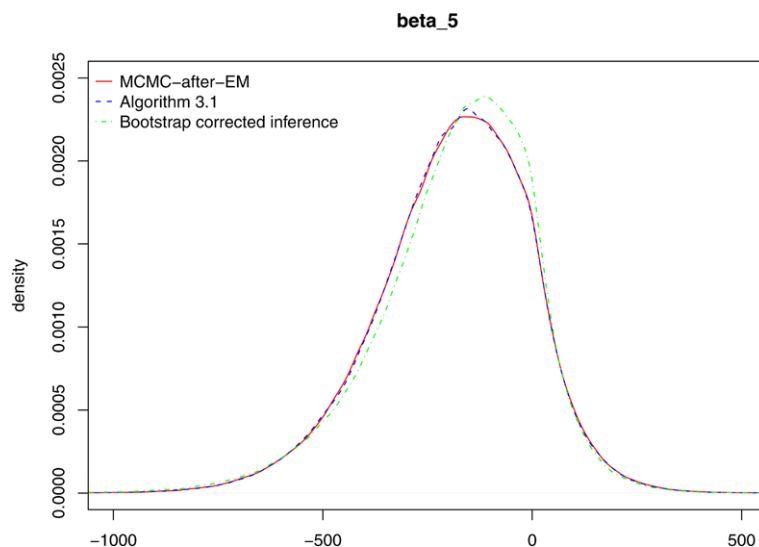


Fig. 2 Diabetes data set. Posterior density estimates for β_5 (regressor *SI*)



This integral is intractable and requires MCMC. Thus, given λ_n , a full MCMC sampler with target distribution $\pi(\theta|y, \lambda_n)$ is run until convergence. The output of this MCMC sampler is used to approximate λ_{n+1} .

We compare these two strategies using the diabetes data used in Park and Casella (2008). This data set has $n = 442$ and $p = 10$. For the EM algorithm, we start from $\lambda_0 = 1$ and perform 30 iterations of the EM. For each such iteration, we run the Gibbs sampler for 10,000 iterations in order to estimate the Q function. For Algorithm 3.1, we use the stabilization described in Algorithm 2.2. We use the same starting point $\lambda_0 = 1$ and the sampler is run for 10,000 iterations. In the implementation of Algorithm 3.1, we use $a_n = 1/n$

and use compact sets of the form $K_n = [-n - 1, n + 1]$ to control λ_n .

Figure 1 (a) shows the sequence e^{λ_n} from the 30 iterations of the EM algorithm. This sequence converges towards 0.237, the same value obtained in Park and Casella (2008). Figure 1 (b) plots the trace plot of e^{λ_n} from Algorithm 3.1 which has also settled around 0.237. Figure 1 (c) reports the empirical marginal distribution of $e^{\hat{\lambda}}$ obtained by the bootstrap procedure described in Sect. 2.3 with $K = 100$.

We then compare the outputs from the three methods through their estimates of the posterior densities of the parameters. Figure 2 shows the estimated posterior densities for the coefficient of the regressor *SI* (for which the differences between the methods are the most noticeable). The densi-

ties are estimated from 500,000 iterations of each algorithm. We can observe from that figure that MCMC-after-EM and Algorithm 3.1 produce virtually the same posterior distribution. The posterior density estimate from the bootstrap-corrected empirical Bayes analysis differs slightly but not by much given the range of the variable. We are therefore inclined to conclude that the Bayesian LASSO model in this example is not particularly sensitive to the choice of λ in the vicinity of the maximum likelihood estimate.

3.2 Bayesian variable selection

Variable selection plays an important role in knowledge discovery. In an influential paper George and Foster (2000) proposed an empirical Bayes variable selection method for linear models. But due to computational difficulties, the paper falls short in implementing their proposed marginal maximum likelihood empirical Bayes method. We show in this example that our framework provides a straightforward implementation of that method.

Let $y \in \mathbb{R}^n$ be the response vector, X a $n \times p$ matrix of regressors, the columns of which are denoted by X_i , $i \in \{1, \dots, p\}$. We are interested in linear regression models of y over subsets of $(X_1, \dots, X_p)$. We index such models by $\gamma \in \{0, 1\}^p$. For a given model $\gamma \in \{0, 1\}^p$, $\gamma_i = 1$ means that the regressor X_i is included in model γ and $\gamma_i = 0$ means that the regressor is excluded from the model. The quantity $q_\gamma = \sum_{i=1}^p \gamma_i$ is the size of the model. We write X_γ to denote the $n \times q_\gamma$ matrix of regressors included in model γ . If model γ hold, then for some vector of parameter β_γ , we have

$$y = X_\gamma \beta_\gamma + \varepsilon,$$

where $\varepsilon \sim \mathcal{N}(0, \sigma^2 I_n)$, where $\mathcal{N}(\mu, \Sigma)$ denote the multivariate normal distribution with mean μ and covariance matrix Σ . We assume a G-prior for β_γ :

$$\beta_\gamma \sim \mathcal{N}(0, e^c \sigma^2 (X_\gamma' X_\gamma)^{-1}),$$

for some hyper-parameter $c \in \mathbb{R}$.

A common practice that we follow here is to assume the improper prior distribution $1/\sigma^2$ for the parameter σ^2 . Although this prior distribution is improper, it is known to yield a proper posterior distribution. One of the main reason for the popularity of g-priors in linear regression is tractability. Indeed we can write out the joint conditional density of $(y, \beta_\gamma, \sigma^2)$ given (γ, c) and integrate out β_γ and σ^2 to obtain the distribution of y given γ and c :

$$f_{\gamma, c}(y) \propto \frac{S(c, \gamma)^{-n/2}}{(1 + e^c)^{q_\gamma/2}}, \quad (9)$$

where

$$S(c, \gamma) = y' y - \frac{e^c}{1 + e^c} \hat{\beta}_\gamma' X_\gamma' X_\gamma \hat{\beta}_\gamma.$$

In the above, $\hat{\beta}_\gamma$ denotes the usual least squares estimate of β_γ in model γ . It is important to mention that the normalizing constant referred to in (9) does not depend on (γ, c) .

We assume independent Bernoulli prior for γ . That is, for some hyper-parameter $\omega \in \mathbb{R}$, $\pi(\gamma|\omega) = F(\omega)^{q_\gamma} (1 - F(\omega))^{p-q_\gamma}$, where $F(\omega) = e^\omega / (1 + e^\omega)$. This leads to the posterior distribution of γ given (y, c, ω) :

$$\pi(\gamma|y, c, \omega) \propto F(\omega)^{q_\gamma} (1 - F(\omega))^{p-q_\gamma} \frac{S(c, \gamma)^{-n/2}}{(1 + e^c)^{q_\gamma/2}}. \quad (10)$$

The marginal maximum likelihood empirical Bayes method of George and Foster (2000) consists in sampling from the posterior distribution $\pi(\gamma|y, c, \omega)$ where the hyper-parameter (c, ω) are set to their maximum likelihood estimate obtained by maximizing the log-likelihood $\ell(c, \omega) \stackrel{\text{def}}{=} \log \pi(y|c, \omega)$ where $\pi(y|c, \omega)$ is given by

$$\pi(y|c, \omega) \propto \sum_{\gamma} F(\omega)^{q_\gamma} (1 - F(\omega))^{p-q_\gamma} \frac{S(c, \gamma)^{-n/2}}{(1 + e^c)^{q_\gamma/2}}.$$

The summation over all models γ makes the direct maximization of this likelihood intractable. We now apply Algorithm 2.1 as detailed in Sect. 2. The function H here takes the form

$$H(c, \omega; \gamma) = \left(\frac{ne^c}{2(1+e^c)^2} \frac{\hat{\beta}_\gamma' X_\gamma' X_\gamma \hat{\beta}_\gamma}{S(c, \gamma)} - \frac{q_\gamma e^c}{2(1+e^c)} \right) / (q_\gamma - pF(\omega)).$$

We need one last ingredient. Given (c, ω) , we need a Markov kernel with invariant distribution $\pi(\gamma|y, c, \omega)$. This can be done very easily using the Metropolis-Hastings algorithm: we randomly select an index $i \in \{1, \dots, p\}$ and flip γ_i to $1 - \gamma_i$. The move is then accepted or rejected with the appropriate Metropolis acceptance ratio. Let us call $P_{c, \omega}$ this Markov kernel. We are now in position to implement Algorithm 2.1.

Algorithm 3.2

- (i) *Initialize* $\gamma_0 \in \{0, 1\}^p$, $(c_0, \omega_0) \in \mathbb{R} \times \mathbb{R}$ arbitrarily. We use $\gamma_0 \equiv 1$, $c_0 = 5$ and $\omega_0 = 0$. Let $\{a_k, k \geq 0\}$ be a positive sequence such that $a_k \rightarrow 0$ and $\sum a_k = \infty$. We use $a_n = 1/n^{0.8}$.
- (ii) *The recursion is the following. At time k , given $(c_k, \omega_k, \gamma_k)$:*
 1. *Sample* $\gamma_{k+1} \sim P_{c_k, \omega_k}(\gamma_k, \cdot)$, where $P_{c, \omega}$ is the Metropolis kernel described above.

2. Calculate

$$c_{k+1} = c_k + a_k \left(\frac{ne^{c_k}}{2(1+e^{c_k})^2} \frac{\hat{\beta}'_{\gamma_{k+1}} X'_{\gamma_{k+1}} X_{\gamma_{k+1}} \hat{\beta}_{\gamma_{k+1}}}{S(c_k, \gamma_{k+1})} - \frac{q_{\gamma_{k+1}} e^{c_k}}{2(1+e^{c_k})} \right),$$

$$\omega_{k+1} = \omega_k + a_k (q_{\gamma_{k+1}} - pF(\omega_k)).$$

In practice we implement the algorithm using the stabilization technique described in Algorithm 2.2. As explained above, Algorithm 3.2 generates a stochastic process $\{(\gamma_k, c_k, \omega_k), k \geq 0\}$ where (c_k, ω_k) converges towards the maximum likelihood estimate (c_*, ω_*) of (c, ω) and where the distribution of γ_k converges towards $\pi(\gamma|y, c_*, \omega_*)$.

The sequence $\{\gamma_k, k \geq 0\}$ can be used to perform the actual variable selection procedure. One standard approach to that end consists of choosing along the chain $\{\gamma_k, k \geq 0\}$ the model γ_k for which $\pi(\gamma_k|y, c_k, \omega_k)/\pi(\bar{\gamma}|c_k, \gamma_k)$ is maximum, where $\pi(\cdot|y, c_k, \omega_k)$ is the posterior distribution given in (10) and $\bar{\gamma}$ some arbitrary model, for example $\bar{\gamma} = \gamma_0$.

We now illustrate the algorithm with the following simulated example adapted from George and Foster (2000). We take $p = 50$ variables and $n = 200$ cases. With $\sigma = 1$, we generate the data (y, X) as follows. The n rows of the matrix X are independently generated from $\mathcal{N}_p(0, \Sigma)$ where the ij -th element of Σ is $\rho^{|i-j|}$ with $\rho = 0.8$. In choosing β , we consider two cases. In the first case, 10% of the components of β are non-zero, whereas in the second case 90% of the components of β are non-zero. Given X and β , we generate $y \sim \mathcal{N}_n(X\beta, \sigma^2 I_n)$.

For each data set, we ran Algorithm 3.2 a hundred (100) times. During each run we perform the variable selection procedure described above. Since we know the true model we can compare it to the selected model. We do so by computing the proportion of incorrect inclusions in and exclusions from the selected model

$$L_1(\hat{\gamma}) = \frac{1}{p} \sum_{i=1}^p |\gamma_i - \hat{\gamma}_i|,$$

where in the above, γ denotes the correct model. We also compute the quantity

$$L_2(\hat{\gamma}, \hat{\beta}) = \frac{1}{q_{\hat{\gamma}}} \|X\beta_* - X_{\hat{\gamma}}\hat{\beta}\|^2,$$

where β_* is the true value of β . If $\hat{\gamma} = \gamma$ then $L_1(\hat{\gamma}, \hat{\beta}) \approx \|H_{\gamma}\varepsilon\|^2/q_* \sim \sigma^2 \chi_{q_*}^2/q_*$, where $H_{\gamma} = X_{\gamma}(X'_{\gamma}X_{\gamma})^{-1}X'_{\gamma}$ and q_* is the number of nonzero elements of β_* . Therefore we expect $L_1(\hat{\gamma}, \hat{\beta}) \approx \sigma^2 = 1$. We average these two statistics over the 100 replications. The results are shown in Table 1. We see that with high probability Algorithm 3.2 can successfully recover the correct model. We also report

Table 1 Average of $L_1(\hat{\gamma})$ and $L_2(\hat{\gamma}, \hat{\beta})$ over 100 replications of Algorithm 3.2 for each data set

% of non-zero terms	10%	90%
$\mathbb{E}(L_1(\hat{\gamma}))$	0.009	0.053
$\mathbb{E}(L_2(\hat{\gamma}, \hat{\beta}))$	1.088	1.085

in Fig. 3 the trace plots of the hyper-parameters e^c and $F(\omega)$ over one run of the algorithm. In this example we have used the step size $a_n = 1/n^{0.8}$ compared $a_n = 1/n$ in the previous example. As a result we have more variability in estimating (c_*, ω_*) as one can see from Fig. 3. But in return this choice leads to a better behavior in term of bias.

4 Convergence

We give here a number of sufficient conditions under which Algorithm 2.2 can be shown to converge to the right limit. All the technical tools needed can be found in greater generality in Andrieu et al. (2005) and Atchade et al. (2009). We need two types of conditions. We need Lyapunov-type conditions on the function h (A1) and we need some additional assumptions on the convergence properties of the Markov kernels P_{λ} . Let us denote $\mathbb{P}_{\theta_0, \lambda_0}^{(m)}$ and $\mathbb{E}_{\theta_0, \lambda_0}^{(m)}$ (resp. $\check{\mathbb{P}}_{\theta_0, \lambda_0, m, l}$ and $\check{\mathbb{E}}_{\theta_0, \lambda_0, m, l}$) the probability distribution and expectation operator of the random process generated by Algorithm 2.1 when the sequence of step-size is $\{a_{m+k}, k \geq 0\}$. (resp. Algorithm 2.2 when the sequence of step-size is $\{a_{m+l+k}, k \geq 0\}$ and the initial compact is K_m) on the canonical space $(\Theta \times \Lambda)^{\infty}$.

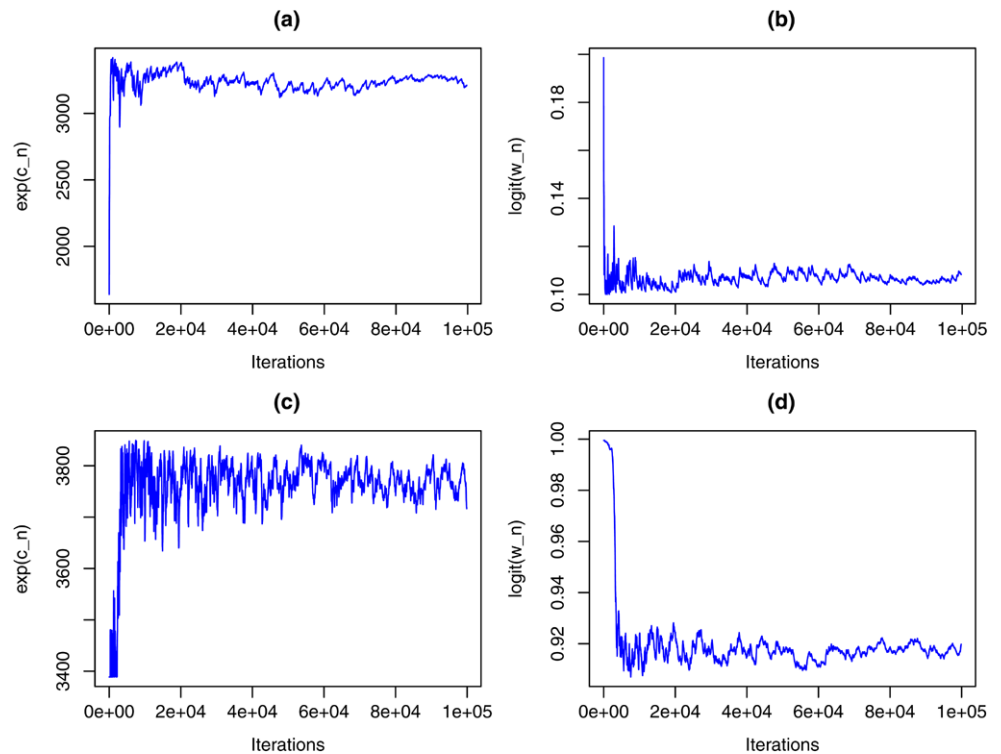
- A1** (i) The set $S \stackrel{\text{def}}{=} \{\lambda \in \Lambda : h(\lambda|y) = 0\}$ is nonempty and finite.
- (ii) The function $\ell(\cdot|y)$ is continuously differentiable and there exists $C \in (0, \infty)$ such that $\ell(\lambda|y) \leq C$ for all $\lambda \in \Lambda$ and $y \in Y$.
- (iii) There exists $M_0 \in \mathbb{R}$ such that for any $M \geq M_0$, the set $\{\lambda \in \Lambda : \ell(\lambda|y) \geq M\}$ is a compact subset of $\mathbb{R}^{n_{\lambda}}$ for all $y \in Y$.

If $W : \Theta \rightarrow [1, +\infty)$ is a function, the W -norm of a function $f : \Theta \rightarrow \mathbb{R}$ is defined as $|f|_W \stackrel{\text{def}}{=} \sup_{\theta \in \Theta} |f(\theta)|/W(\theta)$. The set of functions with finite W -norm is denoted by $\mathcal{L}_W$. Also, for $\lambda, \lambda' \in \Lambda$, define

$$D_W(\lambda, \lambda') = \sup_{|g|_W \leq 1} \sup_{\theta \in \Theta} \frac{|P_{\lambda}g(\theta) - P_{\lambda'}g(\theta)|}{W(\theta)}.$$

When W is the constant function $W \equiv 1$, we write $D(\lambda, \lambda')$. On the Markov kernels, we impose the following condition.

Fig. 3 Sample paths of $\{e^{c_k}, k \geq 0\}$ and $\{1/(1 + e^{-\omega_k}), k \geq 0\}$ from 100,000 iterations of Algorithm 3.2. (a)–(b) (resp. (c)–(d)) corresponds to the data set with 10% (resp. 90%) of nonzero coefficients in true β



A2 Θ is equipped with a countably generated σ -algebra $\mathcal{B}$ and there exist $p \geq 2$, probability measures ν, π_λ on Θ ; a measurable function $V : \Theta \rightarrow [1, \infty)$, a set $\mathcal{C} \in \mathcal{B}$, $\nu(\mathcal{C}) > 0$ with respect to which the following hold.

- (i) P_λ has invariant distribution π_λ and for any compact $K \subset \Lambda$, there exists $\varepsilon > 0$ such that for any $(\theta, \lambda) \in \Theta \times K$, $P_\lambda(\theta, \cdot) \geq \mathbf{1}_{\mathcal{C}}(\theta)\varepsilon\nu(\cdot)$.
- (ii) For any compact $K \subset \Lambda$, there exist constants $\rho \in (0, 1)$, $b \in (0, \infty)$ such that for any $(\theta, \lambda) \in \Theta \times K$, $P_\lambda V(\theta) \leq \rho V(\theta) + b\mathbf{1}_{\mathcal{C}}(\theta)$.
- (iii) For any compact $K \subset \Lambda$, there exists a constant C such that for any $\lambda, \lambda' \in K$,

$$D(\lambda, \lambda') + D_{V^{1/p}}(\lambda, \lambda') + D_V(\lambda, \lambda') \leq C|\lambda - \lambda'|. \quad (11)$$

A3 There exists $0 \leq \beta \leq 1/p$ (where p is as in A2) such that for any compact $K \subset \Lambda$ and $\lambda, \lambda' \in K$

$$\sup_{\lambda \in K} \sup_{\theta \in \Theta} \frac{|H(\theta, \lambda)|}{V^\beta(\theta)} < \infty, \\ \sup_{\theta \in \Theta} \frac{|H(\theta, \lambda) - H(\theta, \lambda')|}{V^\beta(\theta)} \leq C|\lambda - \lambda'|.$$

Theorem 4.1 Assume A1–A3, (5) and suppose that $\Lambda_0 \subseteq K_0$ and $\sup_{\theta_0} V(\theta) < \infty$. Then $\{\lambda_n, n \geq 0\}$ remains almost surely in a compact set and there exists a $\mathcal{S}$ -valued random variable $\lambda_\star$ such that λ_n converges almost surely to $\lambda_\star$.

Moreover

$$\lim_{n \rightarrow \infty} \sup_{|f| \leq 1} |\check{\mathbb{E}}_{\theta_0, \lambda_0}(f(\theta_n) - \pi_{\lambda_\star}(f))| = 0. \quad (12)$$

Proof See Appendix. $\square$

Remark 4.1 Theorem 4.1 gives a set of sufficient conditions under which Algorithm 2.2 is guaranteed to converge. We briefly comment on these assumptions and discussed their applicability to the examples presented above.

In A1 we require that the log-likelihood function be smooth (continuously differentiable), bounded from above, with a finite number of local modes and with compact upper level sets. This is a fairly natural set of conditions to impose when dealing with likelihood maximization problems. For example Lange (1995) made similar assumptions in the convergence analysis of the EM algorithm. Notice that the compactness of level sets and the finiteness of $\mathcal{S}$ will follow from the smoothness and the boundedness assumptions if in addition we impose that $\lim_{\|\lambda\| \rightarrow \infty} \ell(\lambda|y) = -\infty$. In both examples presented above, one can easily check that the log-likelihood function ℓ is indeed smooth, bounded from above and $\lim_{\|\lambda\| \rightarrow \infty} \ell(\lambda|y) = -\infty$. That is, A1 hold.

We assume in (A2) that the Markov kernels P_λ are geometrically ergodic and Lipschitz in λ . This type of conditions have been considered by many authors in the analysis of adaptive MCMC algorithms (see Atchade et al. 2009 and the references therein). The Lipschitz condition (A2(iii))

is easy to check and is known to hold for many standard MCMC kernels (e.g. Metropolis algorithms or Gibbs sampler). The most difficult condition here is the Foster-Lyapunov drift condition A2(ii). But this is not specific to the present algorithm: checking which MCMC kernels are geometrically ergodic and which are not is a well known difficult problem. Again we refer the reader to Atchade et al. (2009) (for some pointers to the literature). We should mention that geometric ergodicity (that is A2(ii)) is not really essential to the convergence of Algorithm 2.2. It should be possible to prove a result similar to Theorem 4.1 with A2(ii) replaced by a polynomial drift condition. Perhaps using arguments similar to those developed in Atchade and Fort (2010). But we do not pursue this here.

In the case of Example 3.1, we do not know whether A2(ii) hold or not. Things are nicer in the case of Example 3.2: the state space is finite and the Metropolis kernel P_λ is smooth in λ thus A2 trivially hold.

Once the Foster-Lyapunov drift condition A2(ii) is established, A3 is usually straightforward to check. For example in Example 3.2, $V \equiv 1$ and since $\Theta = \{0, 1\}^p$ is finite, A3 is easily shown.

5 Concluding remarks

This paper has presented an algorithm to handle adaptively and automatically the estimation of hyper-parameter in empirical Bayes inference. The algorithm has the same computational complexity as a standard MCMC algorithm to sample from the posterior distribution with the hyper-parameter held fixed. We have used two examples to show that the algorithm is easy implement and behaves very well in practice. We have established a formal proof of convergence. One possible direction for future research is to investigate the properties of this algorithm in large scale applications, particularly in the small n large p paradigm.

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Appendix: Proof of Theorem 4.1

Define $w(\lambda) = -l(\lambda|y) + C$, where C is the constant in A1(ii). Then $w(\lambda) \geq 0$ for all $\lambda \in \Lambda$ and $\langle \nabla w(\lambda), h(\lambda) \rangle = -\|h(\lambda)\|^2 \leq 0$. This means that the function w is a Lyapunov function for the SA algorithm. The reader can then easily check that A1 above implies assumption A1 of Andrieu et al. (2005). Assumption A2–A3 above is the same as Assumption (DRI) of Andrieu et al. (2005). Then by Theorem 5.5 of Andrieu et al. (2005), we conclude that there exists a $\mathcal{S}$ -valued random variable $\lambda_\star$ such that λ_n converges almost surely to $\lambda_\star$.

Using A2, we can apply Theorem 1.3.2 of Atchade et al. (2009) which states that for any $\epsilon > 0$, there exists $n_0, N, n_0 \geq N$ such that for all $n \geq n_0$,

$$\sup_{|f| \leq 1} |\check{\mathbb{E}}_{\theta_0, \lambda_0}(f(\theta_n) - \pi_{\lambda_{n-N}}(f))| < \epsilon. \quad (13)$$

Therefore for $n \geq n_0$,

$$\begin{aligned} & \sup_{|f| \leq 1} |\check{\mathbb{E}}_{\theta_0, \lambda_0}(f(\theta_n) - \pi_{\lambda_\star}(f))| \\ & \leq \sup_{|f| \leq 1} |\check{\mathbb{E}}_{\theta_0, \lambda_0}(f(\theta_n) - \pi_{\lambda_{n-N}}(f))| \\ & \quad + \sup_{|f| \leq 1} |\check{\mathbb{E}}_{\theta_0, \lambda_0}(\pi_{\lambda_{n-N}}(f) - \pi_{\lambda_\star}(f))| \\ & \leq \epsilon + \check{\mathbb{E}}_{\theta_0, \lambda_0} \left[\sup_{|f| \leq 1} |\pi_{\lambda_{n-N}}(f) - \pi_{\lambda_\star}(f)| \right]. \end{aligned}$$

The assumption that the σ -algebra is countably generated guarantees that $\lambda \rightarrow \sup_{|f| \leq 1} |\pi_\lambda(f)|$ is measurable (see e.g. Roberts and Rosenthal 1997 Proposition 4.1 for a proof). By the dominated convergence theorem, the proof will be finished if we show that $\lim_{n \rightarrow \infty} \sup_{|f| \leq 1} |\pi_{\lambda_n}(f) - \pi_{\lambda_\star}(f)| = 0$ almost surely.

Fix $\bar{\theta} \in \Theta$ arbitrary. Consider a sample path for which there exists a compact set K such that $\lambda_n \in K$ for all $n \geq 0$ and such that $\lim_{n \rightarrow \infty} \lambda_n = \lambda_\star$. For any $r \geq 1$ and on such sample path, we have $|\pi_{\lambda_n}(f) - \pi_{\lambda_\star}(f)| \leq |\pi_{\lambda_n}(f) - P_{\lambda_n}^r f(\bar{\theta})| + |P_{\lambda_n}^r f(\bar{\theta}) - P_{\lambda_\star}^r f(\bar{\theta})| + |P_{\lambda_\star}^r f(\bar{\theta}) - \pi_{\lambda_\star}(f)|$. Assumption A2 (applied with the path-dependent compact K) implies that there exist a finite constant C and $\rho_0 \in (0, 1)$ (these constants depend on the chosen sample path) such that for all $r \geq 1$

$$|\pi_{\lambda_n}(f) - P_{\lambda_n}^r f(\bar{\theta})| + |P_{\lambda_\star}^r f(\bar{\theta}) - \pi_{\lambda_\star}(f)| \leq CV(\bar{\theta})\rho_0^r.$$

Since $P_{\lambda}^r f(\theta) - P_{\lambda'}^r f(\theta) = \sum_{j=0}^{r-1} P_{\lambda}^j (P_{\lambda} - P_{\lambda'}) (P_{\lambda'}^{k-j-1} - \pi_{\lambda'}) f(\theta)$, we deduce that on the same sample path, we have $|P_{\lambda_n}^r f(\bar{\theta}) - P_{\lambda_\star}^r f(\bar{\theta})| \leq CV(\bar{\theta})|\theta_n - \theta_\star| \sum_{j=0}^{r-1} \rho_0^j \leq CV(\bar{\theta})(1 - \rho_0)^{-1}|\theta_n - \theta_\star|$. It follows that for all $r \geq 1$ and $n \geq 1$,

$$\begin{aligned} \sup_{|f| \leq 1} |\pi_{\lambda_n}(f) - \pi_{\lambda_\star}(f)| & \leq CV(\bar{\theta})(\rho_0^r + (1 - \rho_0)^{-1} \\ & \quad \times |\lambda_n - \lambda_\star|). \end{aligned}$$

Since r is arbitrary, we let $r \rightarrow \infty$. Taking the limit in n , we get

$$\lim_{n \rightarrow \infty} \sup_{|f| \leq 1} |\pi_{\lambda_n}(f) - \pi_{\lambda_\star}(f)| = 0.$$

We conclude by noting (as shown above) that for almost all sample, there exists a compact set K such that $\lambda_n \in K$ for all $n \geq 0$ and such that $\lim_{n \rightarrow \infty} \lambda_n = \lambda_\star$.

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